DEEPANKAR DEY

Senior Python Developer | AI & Automation Solutions

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PROFESSIONAL SUMMARY

Python Technical Lead with 8+ years building agentic AI systems, enterprise automation, and document intelligence pipelines at scale. At EXL, led a 6-person team delivering hyper-automation platforms that process 100K+ records annually and OCR pipelines handling 700–800 documents per day. Architected Adjustor.AI — a production-grade, multi-agent RAG system for insurance claims adjudication using GPT-4o-mini, ChromaDB, and real-time agentic workflows. Combines deep Python and Flask engineering with hands-on Generative AI, prompt engineering, and Azure cloud expertise to deliver systems that actually ship.

CORE SKILLS

Programming	Python, SQL, Regex, REST APIs
Backend & APIs	Flask, API design, microservices concepts, JSON integrations
Automation	Selenium, enterprise hyper-automation, workflow orchestration, data pipelines
AI / OCR / NLP	ABBYY OCR, document intelligence, NLP, text extraction, ML-based data processing, LLM, RAG, LangChain, Claude, Prompt Engineering, Agentic AI, Generative AI
Cloud & DevOps	Azure services, AWS (working knowledge), Docker, CI/CD, IIS, Git
Libraries & Tools	Pandas, NumPy, PyMuPDF, multiprocessing, Jira, Agile
Leadership	Technical leadership, mentoring, code reviews, solution design, stakeholder communication

PROFESSIONAL EXPERIENCE

Technical Lead, Python & AI Automation | Cloud Engineer
Nov 2020 – Present

EXL | Noida, Uttar Pradesh

- Engineered Python-based automation workflows using Flask APIs, Azure services, and SQL, automating 16+ processes and reducing manual effort by 30% FTE.
- Led a team of 6 Python developers while remaining hands-on in coding, debugging, and architecture decisions for enterprise automation platforms.
- Implemented scalable automation pipelines processing 100K+ records annually with robust error handling and monitoring.
- Developed OCR and document intelligence pipelines processing 700–800 documents per day, extracting 25+ structured fields using ABBYY OCR and ML techniques.
- Integrated third-party APIs and internal systems to streamline data flow and improve reliability across automation workflows.
- Optimized Azure production environments, deployment pipelines, and security configurations, maintaining >95% security compliance score.
- Mentored engineers through code reviews, design discussions, and best practices for Python development and automation architecture.

Senior Software Engineer / Senior Python Developer
Nov 2017 – Nov 2020

Sopra Steria | Noida, Uttar Pradesh

- Developed Python and Flask-based backend services for enterprise applications supporting automation and ticket intelligence workflows.
- Implemented NLP-based text analysis and classification models to improve ticket resolution accuracy and recommendation quality.
- Designed and integrated REST APIs used across multiple client environments and production deployments.
- Delivered client demos, supported releases, and collaborated with cross-functional teams to enhance solution performance.
- Improved application maintainability by refactoring Python modules and optimizing data-processing logic.

SELECTED PROJECTS

Adjustor.AI — AI-Powered Insurance Claims Platform (EXL) **— Python, Flask, React.js, OpenAI GPT-4o-mini, ChromaDB, RAG, Multi-Agent Orchestration, PyMuPDF**

- Designed and implemented a 5-agent AI pipeline (Classification → Retrieval → PDF Merge → Analysis → Highlight) orchestrated via a custom Python orchestrator with Server-Sent Events (SSE) for real-time streaming to the frontend.
- Built a RAG pipeline using ChromaDB vector store and OpenAI text-embedding-3-small for semantic retrieval of relevant policy clauses from ingested PDF documents.

- Engineered LLM-powered 5-step chain-of-thought reasoning using GPT-4o-mini — covering claim classification, coverage check, exclusion analysis, document hierarchy resolution, and confidence-rated recommendation.
- Implemented agentic email ingestion — autonomous Gmail IMAP polling that classifies, extracts, and persists incoming claims without human intervention.
- Built AI email drafting — post-decision LLM generation of personalised claimant response emails with decision explanation in plain English.
- Developed ambiguity detection — automatic identification of low-confidence signals with AI-generated follow-up questions for adjusters.

Hyper Automation Platform (EXL) **— Python, Selenium, Azure, SQL**

- Automated enterprise workflows using Python and Selenium, enabling scalable execution across multiple business processes.
- Integrated Azure services and APIs to support secure automation and centralized workflow orchestration.
- Enhanced throughput and reliability by implementing modular automation components and retry/error handling mechanisms.

DataCapture Document Intelligence (EXL) **— Python, Flask, OCR, ML**

- Built a document intelligence pipeline converting unstructured PDFs into structured datasets for downstream automation.
- Implemented OCR extraction logic capturing 25+ fields from forms using ABBYY OCR, regex, and ML-assisted techniques.
- Optimized processing performance using multiprocessing to support high-volume document ingestion.

ABBYY Text Extraction (EXL) **— Python, ABBYY FlexiCapture, OCR**

- Engineered OCR-based extraction workflows using ABBYY FlexiCapture and Python to process structured and semi-structured forms.
- Automated extraction of 25+ data fields with approximately 92% accuracy across 250+ document samples.
- Integrated OCR outputs into downstream automation pipelines to reduce manual validation effort.

Smart Ticket Advisory (Sopra Steria) **— Python, Flask, NLP**

- Developed NLP-driven recommendation logic to identify similar incidents and suggest likely resolutions.
- Improved support triage efficiency through automated text analysis and ranking models.

EDUCATION

Bachelor of Technology (Computer Science) 2013 – 2017
ABES Engineering College, Ghaziabad, Uttar Pradesh

Higher Senior Secondary (Science) 2013
St. Paul's Academy, Ghaziabad, Uttar Pradesh

AWARDS & RECOGNITION

★ **Digital Team for the Future Award** (EXL) — Recognised for originating new digital solution pathways.
★ **Einstein Award** (Sopra Steria) — For converting an idea into a production-ready Smart Ticket Advisory solution.
★ **Falcon Award** (Sopra Steria) — For rapid multi-client delivery within tight timelines.

Languages: **English** | **Hindi** | **Bengali**